西南财经大学本科考试

课程论文

学年学期： 2025-2026-2

课程名称： 公司金融（英）

论文题目： Capability Surprise and the Pricing of an Early-Commercial-Stage Foundation-Model Lab: Evidence from Zhipu AI (2513.HK)

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年级专业： 2023 级　金融数学

评语： 得　分：_____ 评阅教师签字：_____　　年　月

Capability Surprise and the Pricing of an Early-Commercial-Stage Foundation-Model Lab: Evidence from Zhipu AI (2513.HK)

GitHub Repository: https://github.com/ericxuzhesheng/Zhipu-AI-Valuation

Abstract

Technical progress can attract a market revaluation before a foundation-model laboratory generates sustainable cash flow. We examine this gap for Zhipu AI using disclosures and prices through 30 September 2026. Scenario DCF, reverse DCF and layered comparables describe the operating outcomes implicit in its valuation; an event study examines returns around five official GLM releases. H1 revenue grew 399.7%, but operating loss reached RMB2.15bn and operating cash outflow RMB3.37bn. The scenario-weighted equity value is HK$116 per share against a HK$641 close, with a substantial contribution from pro-forma net cash. Holding the model's other assumptions fixed, the quoted price requires about US$47.9bn of revenue in 2035. The five releases have a 13.7% mean two-day reaction, rising to 15.3% relative to MiniMax. A block bootstrap supports an unusually large immediate relative response within this sampling design; subsequent drift remains statistically inconclusive and sensitive to the benchmark. Zhipu fell 46.4% during September, although its relative performance improved in the final week. These observations show that the initial re-rating did not provide a durable valuation floor. Overlapping news prevents attribution of the decline to a single cause. Completed financing, ZCode's data-handling controversy and competing developer tools identify commercial channels through which technical capability may affect shareholder cash flow. An illustrative Anthropic IPO model applies the same distinction between revenue expectations and cash-flow value at a different scale. The analysis establishes conditional valuation gaps and short-run responses, rather than a causal test of mispricing or statistical mean reversion.

Keywords: foundation models; Chapter 18C; pre-profit valuation; real options; market efficiency; capability news; commercialisation.

1 Introduction

Zhipu AI (2513.HK) and MiniMax (00100.HK) listed in Hong Kong on 8 and 9 January 2026. Zhipu used Chapter 18C, the regime for specialist-technology firms that admits pre-profit issuers. By 30 September its shares were 5.5 times the IPO price, despite falling from HK$1,195 at the end of August to HK$641. Its post-placement equity value of US$40.1bn represented 191.3× LTM revenue of US$209.5m through 2026H1 [1,3,6,9]. This combination of rapid appreciation, a large subsequent decline and continuing losses raises a valuation question: what must technical progress achieve commercially to support the public-market price?

The paper addresses three related questions. How do investors respond around official model releases? What revenue, margins and investment requirements are consistent with the current price? How do the open-weight strategy, developer adoption and access to capital affect those assumptions? A release can improve a laboratory's technical position without establishing the volume of paid usage, customer retention or eventual cash flow. Financial disclosures remain informative for a loss-making firm, even when earnings multiples offer little guidance.

Model releases provide information events alongside the earnings announcements studied by **(author?)** [10] and **(author?)** [11]. We adapt the reaction and drift windows in the surprise-event design of Guosen Securities [12] to official GLM releases. The empirical variable is the dated announcement: we do not independently measure how far each release exceeded investors' prior expectations. "Capability surprise" in the title denotes this research motivation, rather than an estimated surprise score. Combining return patterns with scenario and reverse DCF connects the market response to technical news with the commercial outcomes required by the valuation.

All information is restricted to 30 September 2026, including the interim report released after the 24 September close. US daily bars use each exchange's local trading date. The original five-event sample remains fixed at 31 August; September extensions are reported separately. Sections 2 and 3 establish the operating and financing context. Section 4 measures conditional valuation gaps, Section 5 examines the release windows, and Section 6 considers the later repricing and an Anthropic comparison.

2 Background: Company, Financials, and Listing

Founded in 2019 out of Tsinghua University's Knowledge Engineering Group, Zhipu develops the GLM ("General Language Model") family. It earns revenue from usage-based API access, enterprise subscriptions and private or on-premise deployment. Releasing model weights and charging competitive token prices can encourage developer adoption; the commercial objective is to convert that adoption into paid usage through Z.ai and Z.code. Whether this produces attractive unit economics is an empirical question.

The H1 results show progress in adoption alongside substantial operating costs. Revenue rose 399.7% to RMB953.9m; open-platform and API revenue reached RMB825.2m, or 86.5% of the total. API gross margin improved from −0.4% to 24.6%, while the shift toward this business reduced total gross margin from 50.0% to 26.4%. Operating loss widened 13.0% to RMB2.15bn, and adjusted net loss widened 12.1% to RMB1.96bn. Reported net loss narrowed partly because losses on investor financial instruments fell after listing. Underlying losses remained substantial despite the revenue increase[5].

At 30 June the group had positive equity of RMB4.43bn, bank cash and short-term investments of RMB4.50bn, and bank loans plus leases of RMB2.60bn. The formal interim report discloses H1 operating cash outflow of RMB3,374.1m, versus RMB1,327.2m a year earlier. Note 14 identifies RMB106.0m of restricted deposits, leaving cash equivalents of RMB3,887.7m[59]. The August announcement and September report have different information dates even though they describe the same reporting period. Their financial content informs the valuation; the later cash-flow disclosure cannot retrospectively explain earlier returns. Appendix A reproduces the statements and cash reconciliation.

Zhipu is an early-commercial-stage, pre-profit business. Revenue growth, API gross margin, operating expenditure and funding capacity therefore provide more useful anchors than an earnings multiple (Figure 1). Positive API gross margin is a necessary step toward commercial viability, but it does not cover the laboratory's research and other operating costs.

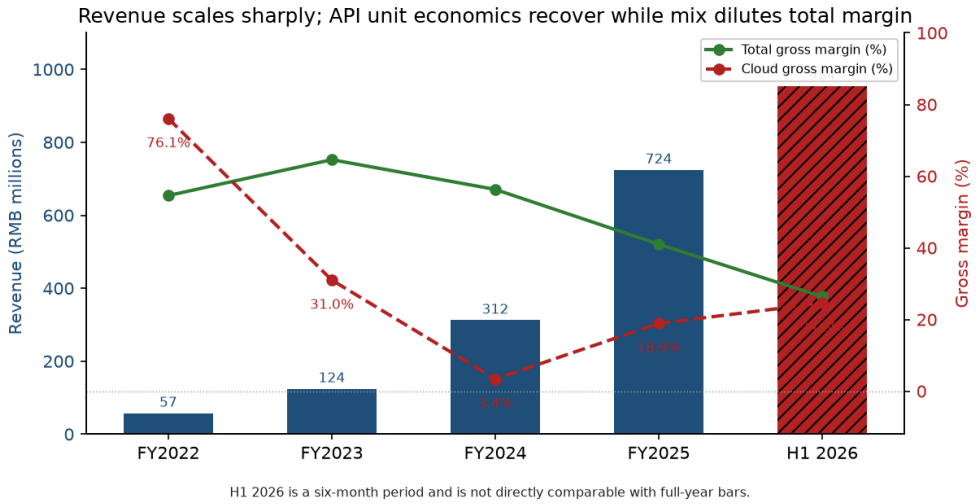


Figure 1: Zhipu's financial profile. H1 2026 is a six-month period and is not directly comparable with the full-year bars. API gross margin recovered to 24.6%, while the shift toward API revenue diluted total gross margin to 26.4%. Sources: Chapter 18C prospectus, 2025 annual report and 2026 interim results[3–5].

Zhipu priced its IPO at HK$116.20, raising about HK$5.0bn after the over-allotment option, with HK$3bn of cornerstone commitments and an implied valuation near US$6.7–7.4bn[3,4]. Chapter 18C provided access to public equity before profitability. The subsequent re-rating also made follow-on financing possible: on 13 July the company placed 19.78m new shares at HK$1,588, raising HK$31.375bn net and increasing issued shares from 445.843m to 465.623m[6]. Further September financing enters the equity bridge in Section 4.

At the September close Zhipu remained approximately 452% above its IPO price, with realised annualised volatility near 184%. MiniMax, which listed at HK$165, closed at HK$250.2. These prices show how substantially market assessments can change during an early public trading history. They do not isolate the effects of model releases, financial results or investor flows.

3 Industry, Macro, and Strategy

The foundation-model business depends on costly compute and competes with both closed and open-weight alternatives. Compute-export restrictions can constrain supply; open weights can make alternative models easier to evaluate. Enterprise integration and data-handling requirements may still make switching costly, so low technical switching barriers should not be equated with low customer switching costs in every deployment.

Zhipu competes on coding performance, model access and price. GLM-4.6 used approximately 355B parameters, 32B active parameters and a 200K context. GLM-5.1 reported a 58.4% SWE-Bench Pro score and a first-place ranking; GLM-5.2 introduced MIT open weights and a one-million-token context in June. GLM-5.3 followed on 14 August and GLM-5.3-Flash on 26 August. The GLM-5.x releases also brought API price increases of 8–17%[13,15,16]. These product facts motivate the dated events in Section 5. They do not independently measure investors' surprise or prove a persistent pricing advantage. Branch releases such as GLM-5V-Turbo are catalogued but excluded when they overlap a main GLM window[14].

The open-weight approach can broaden adoption while also increasing competition from accessible alternatives. Higher token prices, lower inference costs and greater paid volume could improve gross profit, but research spending and compute dependence determine how much reaches operating cash flow. The H1 mix and margin changes show why technical benchmarks

and revenue growth alone cannot settle the valuation question.

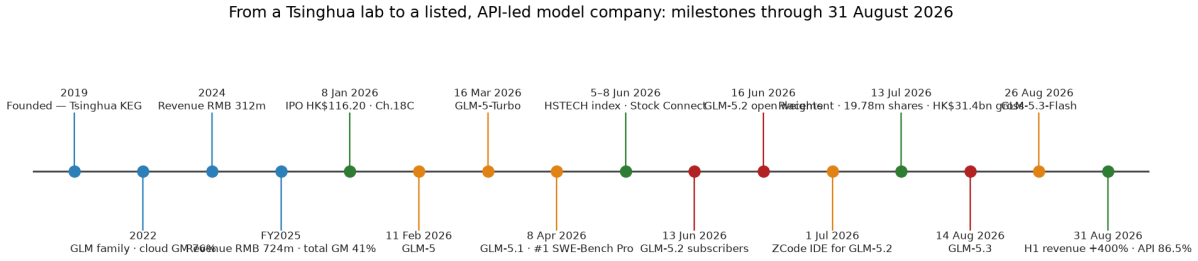


Figure 2: Company timeline from the 2019 founding through the 2026 Chapter 18C listing, July placement, GLM-5.3-Flash rollout and H1 results. Main GLM releases supply the dated information events studied in Section 5; milestones are evenly spaced for legibility. Sources: Chapter 18C prospectus[3], official model documentation[13], placement announcement[6], and interim results[5].

Listed comparables. SenseTime (0020.HK), Fourth Paradigm (6682.HK), and Wenge AI (01956.HK) give local AI context, but their vision, enterprise-ML, and decision-intelligence models differ from a frontier laboratory[35–37]. Moonshot AI (Kimi) and other Chinese labs may eventually broaden the listed set. For now, *MiniMax is the only direct listed comparable* (Section 4.4).

Market access and investor flows. Chinese AI policy and state or SOE demand form part of the demand environment, while compute-export controls constrain available infrastructure. Hong Kong index membership and Stock Connect access also affect the potential investor base. Zhipu joined the Hang Seng Tech Index on 5 June and Southbound Stock Connect on 8 June, with about HK$920m of net southbound buying on its first eligible day[38]. MSCI added its H shares to the Emerging Markets and China indexes at the 31 August close[39]. The event catalogue distinguishes these dates from model and financial announcements. Classification helps interpret the news; it does not prevent their return windows from overlapping.

4 Valuation

4.1 Cost of capital

Zhipu's short post-IPO history makes a regression beta unstable. We therefore estimate 60-month monthly betas for global AI-software comparables against SPY through 31 August, unlever them, and round the median of 1.55 to 1.6[40]. With a 4.0% risk-free rate and 6.0% equity risk premium, the CAPM calculation gives $4.0\% + 1.6 \times 6.0\% = 13.6\%$. We retain 13.5% as the FCFF discount-rate benchmark. This is a modelling convention based on comparable operating risk, not a fully estimated WACC for Zhipu's current capital structure. Mature software peers may understate the risk of a pre-profit laboratory, and the fixed rate does not capture changes in operating or financing risk through 2035. Appendix C records the bridge; sensitivity analysis examines alternative rates.

The model uses HK$7.8/US$, RMB7.1/US$ and 487.588m shares after the September placement. It adds US$6.293bn of pro-forma net cash and short-term investments to operating enterprise value. This bridge combines June cash and investments, less restricted deposits, bank loans and leases, with July and September placement net proceeds and September bond net proceeds, less an issue-date bond-principal proxy[5,6,54]. It is a financing-adjusted bridge, not a reported September cash balance. Unreported H2 operating cash consumption would reduce

the cash available to shareholders.

The 18 September completion announcement confirms that 21,965,000 new H shares were placed at HK$714 on 16 September, raising HK$15,664.13m net. RMB20.14bn of US-dollar-settled zero-coupon convertible bonds were issued on 18 September, raising US$3,010.77m net[54]. The placement's 9.96% discount was measured against the 11 September HK$793 close. Initial conversion is at HK$892.50; maturity is 16 September 2027, and the ordinary conversion period begins on 1 July 2027[52].

The bond claim is represented by US$3,000.64m, calculated as disclosed US$3,015.64m gross proceeds divided by the 100.5% issue price. This proxy is not an IFRS fair-value estimate; dollar settlement depends on the bond terms and exchange rates. We do not separately estimate the conversion option. The non-conversion case uses 487.588m shares and deducts the bond claim. Full conversion at the initial price would increase shares to 513.953m and remove that claim; applying both dilution and the debt deduction would double-count it. Zero coupon still leaves an economic financing cost and a near-term repayment or refinancing obligation. The proposed allocation of 60% of net proceeds to GLM development and infrastructure links the financing to future operating execution.

4.2 Scenario DCF

We project FCFF over 2026–2035 from US$700m of FY2026E revenue. This is an analyst estimate, not company guidance. Starting from reported H1 revenue of RMB953.9m, it implies approximately RMB4.02bn in H2. That increase is demanding, although below the August run-rate reported by management after the results[5,7]. Operating margin begins at −100%, already a sharp improvement from H1's approximately −225%, and rises toward a scenario-specific terminal level. Revenue growth fades from each scenario's 2027 assumption.

The model uses a 15% tax rate, a sales-to-capital ratio of 1.6–2.0 for reinvestment and 4% terminal growth. Negative EBIT generates a simplified NOL carryforward; cash taxes arise only after positive EBIT absorbs it. The loss treatment and terminal extrapolation are simplifying assumptions, rather than a detailed statutory tax forecast. Scenario weights express the analyst's judgement and are not probabilities estimated from market prices. Appendix B gives the Base-case projection and its terminal-value limitation.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD). Market row at the 30 September close. Operating assumptions are unchanged.

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK$)
Bear ($p = 0.35$)	23%	18%	US$3.5B	56
Base ($p = 0.45$)	31%	28%	US$6.3B	102
Bull ($p = 0.20$)	47%	35%	US$15.9B	255
Probability-weighted	—	—	**US$7.3B**	**116**
Market (2026-09-30)	—	—	*US$40.1B*	*641*

The scenario-weighted equity value is HK$116 per share, or 18.1% of the market price. Most of its US$7.256bn equity value comes from the US$6.293bn pro-forma net-cash bridge; the weighted operating enterprise value is approximately US$0.963bn. In the Base case, the operating enterprise value is only about US$55m because the discounted forecast cash deficits nearly offset terminal value. Funding capacity therefore contributes materially to these equity estimates. Raising capital changes the cash and share-count bridge without by itself improving operating profitability.

The difference from market price is conditional on the assumed revenue path, margins, investment requirements, discount rate and financing bridge. These demanding operating assumptions still produce a much smaller equity value than the quote. This result identifies the commercial scale required by the valuation; it does not establish an objective price to which the stock must return.

4.3 Reverse DCF: what the price requires

We invert the model to ask what justifies the US$40.1bn equity value while holding FY2026E revenue fixed. At WACC 13.5% and a 40% terminal margin, the price requires **about US$47.9bn of 2035 revenue**: 2027 growth of roughly 120%, followed by a path that still compounds at 59.9% from 2026 to 2035. Figure 3 grids WACC (10–20%) against terminal margin (20–50%). Even the most favourable cell (10%/50%) requires about US$14.5bn, while 15%/40% requires US$68.1bn. The reverse DCF is therefore a description of the revenue path needed under specified assumptions. Different margins, reinvestment requirements or business models would imply different paths.

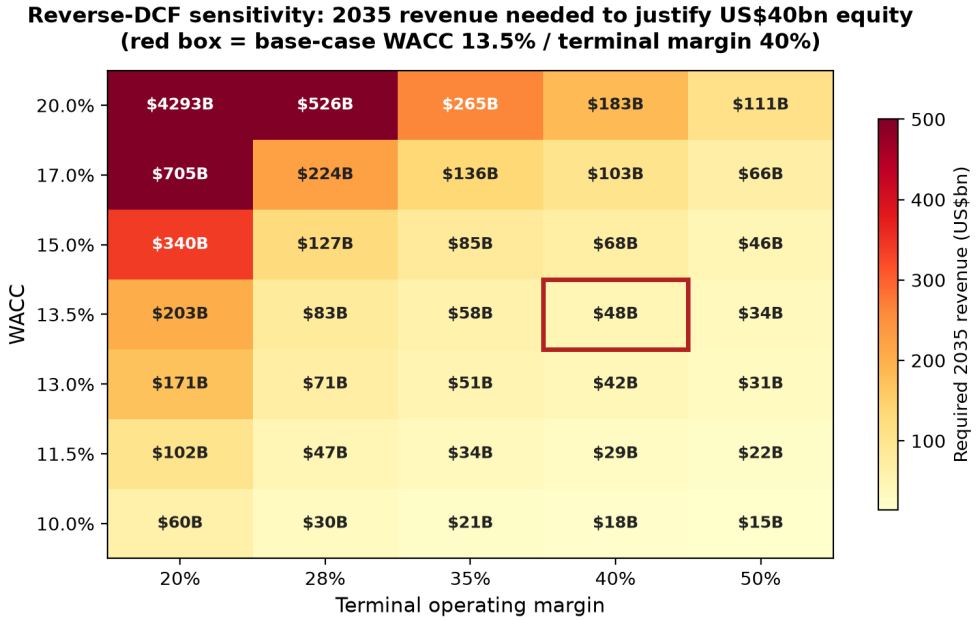


Figure 3: Reverse-DCF sensitivity: 2035 revenue (US$bn) required to justify the US$40.1bn equity value across WACC and terminal-margin assumptions. The red box marks the base case (13.5%/40%). Data: Table 12 in Appendix D.

4.4 Relative valuation

We divide each listed lab's 30 September equity value by LTM revenue through 2026H1. Zhipu's RMB 1,487.3m (FY2025 less H1 2025 plus H1 2026), or US$209.5m at RMB7.1/US$, gives **191.3**×. MiniMax's US$11.20bn equity value and US$165.2m LTM revenue give **67.8**×. Private transactions imply **34.1**× for OpenAI, **20.5**× for Anthropic, and **39.0**× for Mistral[41–44]. This private range, with a 34.1× median, is the football-field transaction range. Applying these transaction multiples to Zhipu's FY2026E revenue estimate gives HK$230–437 per share. MiniMax's LTM multiple applied to Zhipu's LTM revenue gives HK$227.

MiniMax's 26 August results anchor the peer denominator. H1 revenue rose 283.1% to US$116.6m and gross margin reached 17.9%, while adjusted net loss doubled to US$293.0m.

Its US$165.2m LTM revenue equals FY2025 less H1 2025 plus H1 2026, rather than annualised H1 revenue[43].

SenseTime, Phancy, and Wenge are Hong Kong adjacent-AI references; Palantir, Cloudflare, and Snowflake represent commercialisation[45,46]. Different economics make both groups context only. On the matched LTM basis, Zhipu's multiple is 2.82 times MiniMax's (Figure 4).

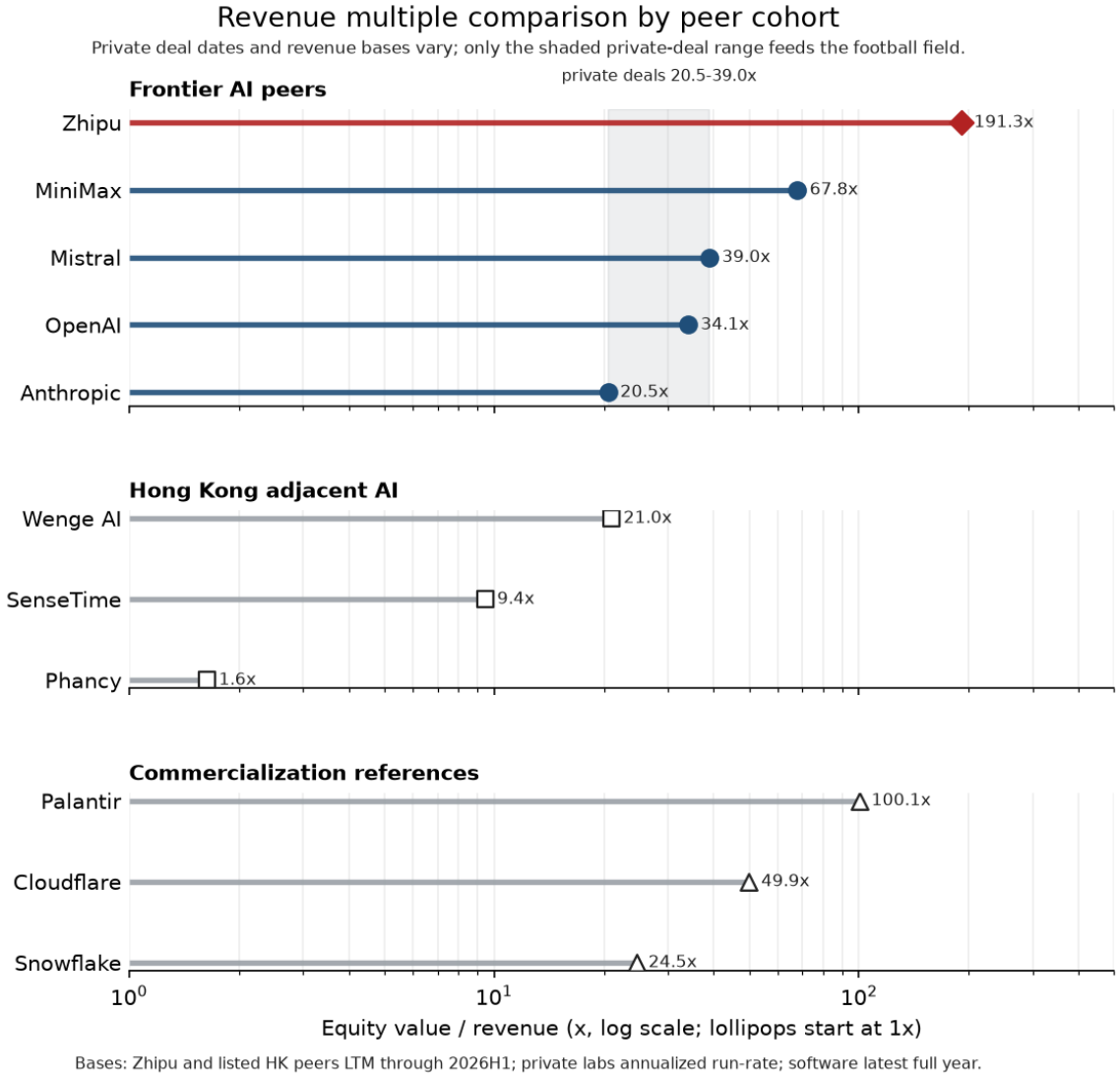


Figure 4: Layered equity value / revenue multiples (log scale). The chart shows core frontier peers, Hong Kong adjacent-AI firms, and commercialisation references in separate groups. Only the shaded private-deal range from 20.5× to 39.0× feeds the football field. Zhipu, MiniMax, and the Hong Kong listed peers use LTM revenue through 2026H1; each other revenue period is recorded in `data/valuation_comps.csv`.

The multiple comparisons require different degrees of caution. MiniMax provides the closest listed laboratory and a matched revenue period, but its product mix, customers and margins still differ. Private financing valuations use transaction-specific terms and revenue run-rates from different dates; applying them to estimated FY2026 revenue is an illustrative conversion, not a matched public-market valuation. The public and private ranges therefore remain separate.

Wenge illustrates the importance of business mix. Its H1 revenue of RMB172.750m was 65.9% deployment and 34.1% subscription and API, with a 55.0% gross margin. Adjusted net loss narrowed 29.5% to RMB52.920m, while R&D expense grew faster than revenue[53]. These eco-

nomics differ from Zhipu's API-heavy mix, so Wenge is an adjacent reference rather than a direct foundation-laboratory valuation anchor. Appendix E records its financial detail and the contribution of fair-value gains to reported losses.

4.5 Sensitivity and synthesis

Figure 5 reports one-at-a-time sensitivity around the Base case. The tested settings produce approximately HK$74–131 per share. None of these individual changes bridges the gap to HK$641, but the test does not cover correlated changes, different business models or higher weights on extreme outcomes.

Figure 6 compares the conditional ranges. Scenario DCF spans HK$56–255; applying the 20.5–39.0× private-deal multiples to FY2026E revenue gives HK$230–437. Applying MiniMax's 67.8× LTM multiple to Zhipu's LTM revenue gives about HK$227, 64.5% below the close. All three approaches leave a material gap, although their denominators and economic assumptions differ. The September decline has therefore reduced the quote without removing the need for substantial future commercial success. Neither HK$116 nor a historical average is an empirically established long-run price mean.

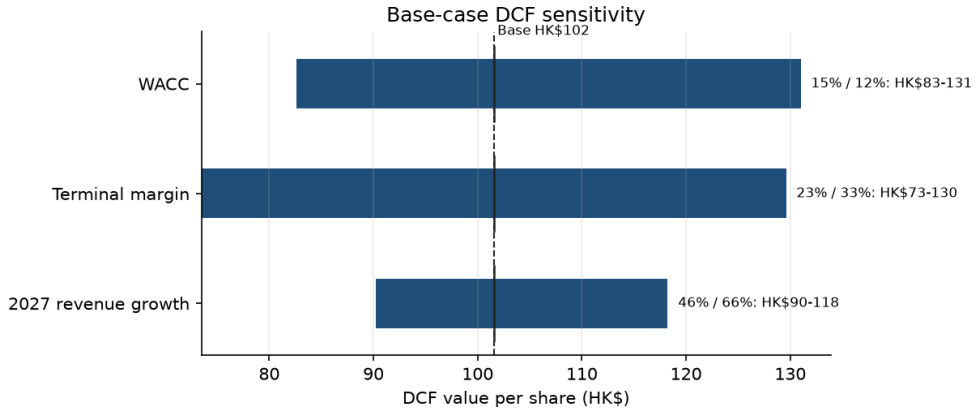


Figure 5: DCF sensitivity (Base case, per share). Bars span each driver's low/high setting around the HK$102 base; no one-at-a-time input change moves value near the market price.

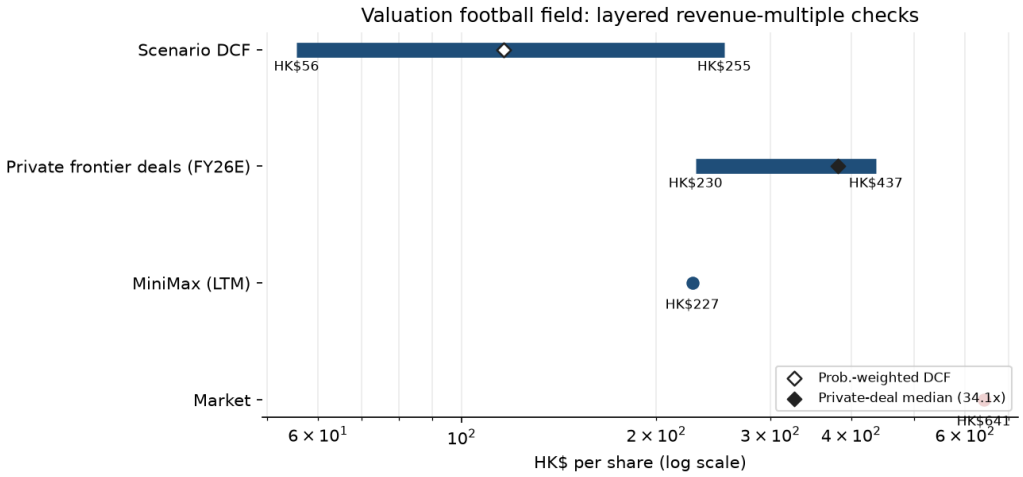


Figure 6: Valuation football field (log scale). The private-deal range and MiniMax's public LTM multiple are shown separately; the diamonds mark probability-weighted DCF and the private-deal median.

5 Capability Surprise: An Event Study

5.1 Design and interpretation

Under semi-strong efficiency, prices incorporate public information promptly[50]. Model announcements may contain information about future paid usage, pricing power or costs. We examine the immediate return response and subsequent drift, drawing on post-earnings-announcement research[10,11]. A positive response is consistent with investors revising their assessment; subsequent continuation or reversal alone does not identify underreaction, overreaction or a change in expected fundamental value.

Event dates come from official release announcements and technical documentation, independently of the price series[13,17]. Daily prices are drawn from Tushare and Yahoo Finance[1,2]. We use leakage $[-5,-1]$, reaction $[0,+1]$ and drift $[+2,+10]$ windows[12,51]. Expected return is Zhipu's mean daily return over the 15-session estimation window $[-20,-6]$, so $AR_t = R_t - \overline{R}_{[-20,-6]}$ and $CAR_{[a,b]} = \sum_{t=a}^{b} AR_t$. CAR is an arithmetic sum of abnormal simple returns, not a compounded buy-and-hold return.

The short estimation window is useful for a recently listed firm but can be unstable during large price moves. Official dates avoid selecting events from price spikes; they do not eliminate anticipation, simultaneous announcements or changes in discount rates. The fixed August cohort and the separate September extension preserve this distinction between event selection and later interpretation.

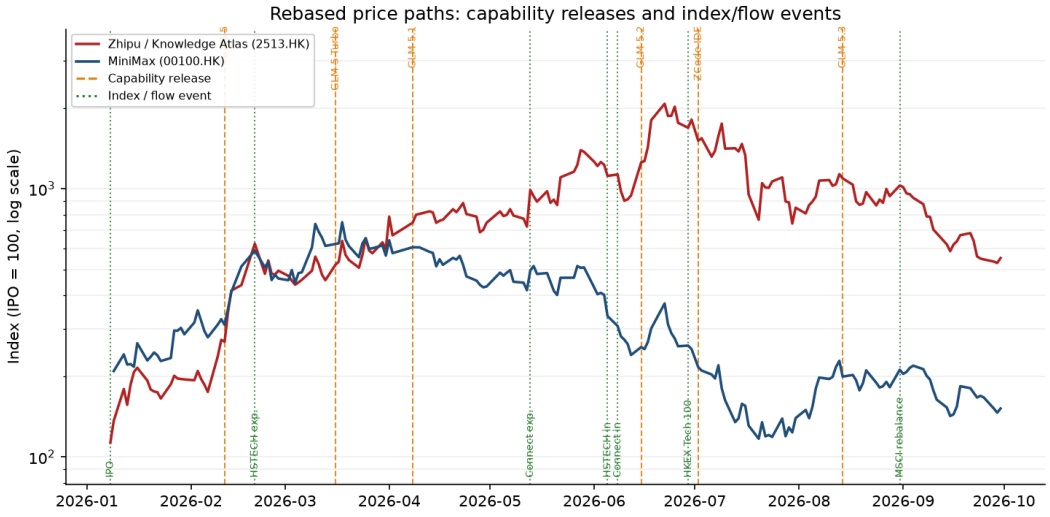


Figure 7: Rebased prices (IPO = 100, log scale, through 2026-09-30). Orange dashed lines mark capability releases (GLM-5 through GLM-5.3); green dotted lines mark index/flow catalysts (IPO, the 20 Feb and 13 May expectation spikes, the 5–8 June Hang Seng TECH inclusion and Stock Connect access, and the 31 August MSCI rebalance). Both stocks decline during September after differing earlier price paths. The chart reports raw prices; it does not decompose the decline into capability, financial or flow effects.

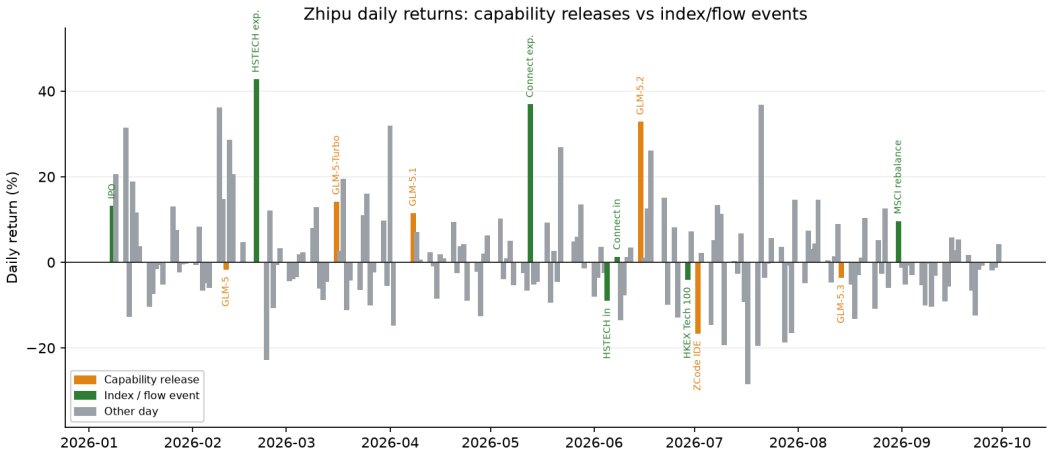


Figure 8: Descriptive cross-check: Zhipu's daily returns with capability-release days (orange) and index/flow days (green) highlighted. Several of the largest up-days coincide with independently dated GLM releases, but the two biggest, +43% on 20 Feb and +37% on 13 May, are classified as index/flow dates rather than model releases. (Event dates are taken from official announcements and index/Connect schedules, not from this chart.)

5.2 Immediate response and subsequent drift

Across the five GLM releases, mean reaction CAR is +13.7% and mean drift is +4.6% (Table 2). Four reactions are positive. Subsequent mean-adjusted drift varies substantially: GLM-5 and GLM-5.2 have positive continuation; GLM-5-Turbo and GLM-5.1 have negative drift; GLM-5.3 has a negative reaction followed by a small positive drift. The release dates therefore coincide with heterogeneous return patterns, rather than a uniform trajectory.

MiniMax's M2.5 has a +24.2% reaction and −2.1% drift, while M2.7 and M3 have negative values in both windows. Large-technology comparators broaden the descriptive context, but their established businesses and different release characteristics prevent treating all these observations as interchangeable laboratory events (Figures 9 and 10).

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %; data through 31 August 2026).

Event	Day 0	Reaction $[0,+1]$	Drift $[+2,+10]$	Reading
GLM-5	2026-02-11	+22.5	+24.7	positive continuation
GLM-5-Turbo	2026-03-16	+7.7	−19.3	negative subsequent CAR
GLM-5.1	2026-04-08	+13.8	−14.2	negative subsequent CAR
GLM-5.2	2026-06-15	+30.8	+28.2	positive continuation
GLM-5.3	2026-08-14	−6.5	+3.8	negative, then positive
MiniMax M2.5	2026-02-12	+24.2	−2.1	positive, then negative
MiniMax M2.7	2026-03-18	−5.5	−49.1	negative in both windows
MiniMax M3	2026-06-01	−21.1	−40.8	negative in both windows
Large-tech AI comparators				
Baidu ERNIE 5.0	2026-02-06	−0.4	−15.8	negative in both windows
Baidu ERNIE 5.1	2026-05-11	−5.0	−13.6	negative in both windows
Alibaba Qwen3.5	2026-02-16	−4.5	−10.9	negative in both windows
Alibaba Qwen3.6	2026-04-16	+6.5	−5.5	negative subsequent CAR
Alibaba Qwen3.7-Max	2026-05-21	−3.4	−2.1	negative in both windows
Alibaba Qwen3.7-Plus	2026-06-01	+8.6	−15.9	negative subsequent CAR
Alibaba Qwen3.8-Max	2026-08-03	+5.3	−12.8	negative subsequent CAR
Tencent Hunyuan 3.0 preview	2026-04-22	−4.2	−1.3	negative in both windows
Tencent Hy3	2026-07-06	+8.2	+9.8	positive continuation
Meituan LongCat-2.0	2026-06-30	+4.8	+17.3	positive continuation
Google Gemini 3.1 Pro	2026-02-19	+4.0	−4.0	negative subsequent CAR
Google Gemini 3.5/Omni	2026-05-19	−4.0	−16.7	negative in both windows
Google Gemini 3.6 Flash	2026-07-21	−2.3	+13.0	negative, then positive
Google Gemini 3.7 Flash	2026-08-13	+0.9	−0.7	muted
Meta Muse Spark	2026-04-08	+11.5	+18.2	positive continuation
Meta Muse Spark 1.1	2026-07-09	+11.1	−7.6	negative subsequent CAR
Average (16 large-tech)		**+2.3**	**−3.0**	
Average (5 GLM)		**+13.7**	**+4.6**	

Mean-adjusted abnormal return: $AR_t = R_t - \bar{R}_{[-20,-6]}$, where $\bar{R}_{[-20,-6]}$ is the average raw return over event days −20 through −6. Comparator dates are pinned to primary-source release announcements or model cards[17–34]. Weekend announcements use the next trading day. OpenAI and xAI releases remain in the audit catalogue, but Microsoft and Tesla returns are excluded because neither listed company is the model issuer. Same-week Qwen3.8 variants are catalogued but not double-counted after the 3 August flagship launch. Qwen3.8-Flash (26 August), Gemini Transcribe/Omni 1.1 (26–27 August), Tencent Hy4 (28 August), GLM-5.3-Flash, the two H1 disclosures, and MSCI rebalance lacked a complete post-event window at the original 31 August sample cutoff and remain outside this historical table. Later observable windows are analysed below. MiniMax H3 and Music 3.0 remain catalogued but are excluded because audio-video and music generation are not comparable with the text, coding, and agent capability sample.

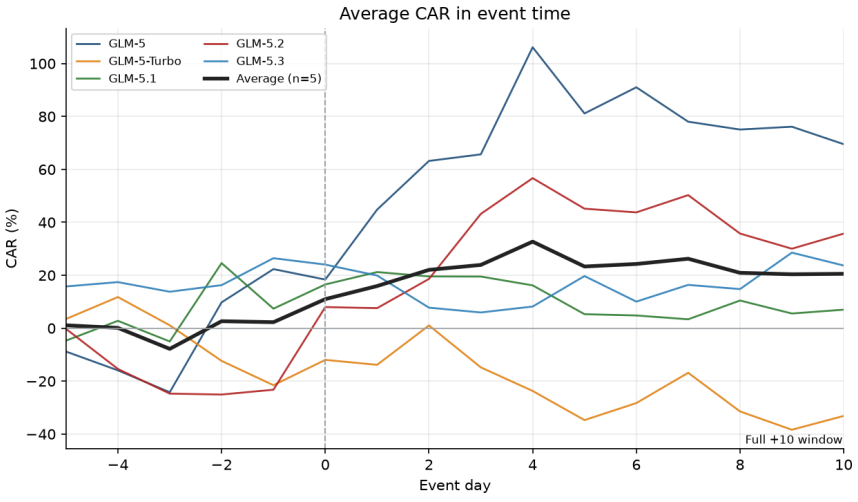


Figure 9: Average CAR in event time across the five GLM events (mean-adjusted, $[-20,-6]$ estimation window). The average is plotted where all five GLM events have data through the full $[+2,+10]$ drift window.

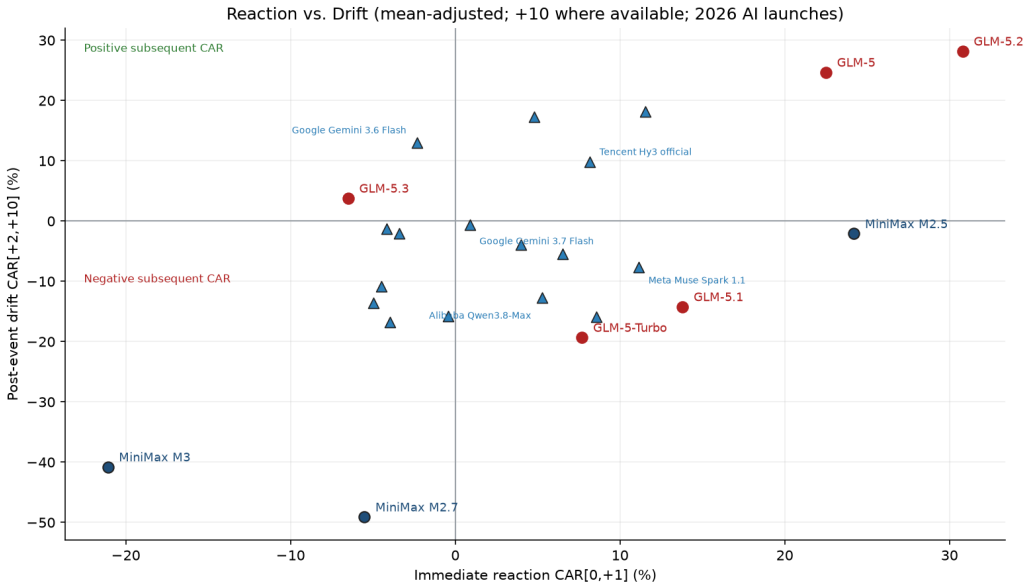


Figure 10: Reaction versus drift under the mean-adjusted benchmark. The quadrants describe the signs of the two CARs; they do not independently identify pricing errors. Triangle markers show large-tech AI comparators.

5.3 Overlapping news and the peer benchmark

Two large raw up-days, approximately +43% on 20 February and +37% on 13 May, are catalogued as index or flow dates. The February observation also falls inside GLM-5's drift window. Index membership and Stock Connect news can therefore contribute to CARs around a capability announcement; categorising the initiating event does not produce an isolated capability effect. Zhipu's 9.6% rise on 31 August preceded the after-close H1 results and coincided with the effective MSCI rebalance[5,39]. The H1 disclosure cannot explain a return that occurred before its release.

The expanded catalogue records model releases, financial disclosures, flow catalysts, listings and screened candidates, with reasons for exclusions. Twenty-nine events have complete windows in the historical CAR panel. "Complete" describes available observations, not the absence of other news. This catalogue makes sample construction auditable without eliminating confounding events.

We also calculate $AR_t = R_{Zhipu,t} - R_{MiniMax,t}$. With this peer benchmark, the five-event mean reaction rises from +13.7% to +15.3%, and mean drift from +4.6% to +16.8% (Table 3). All five relative drift estimates are positive. GLM-5-Turbo changes from −19.3% to +19.2%, and GLM-5.1 from −14.2% to +12.9%. The sign changes show why the mean-adjusted reversals should not be labelled evidence of overreaction.

MiniMax shares some sector and macro exposures with Zhipu, but also has its own product announcements: M2.5 followed GLM-5 by one day, and M2.7 followed GLM-5-Turbo by two. The adjusted estimates measure relative performance against this particular firm, rather than removing all sector shocks. They support relative strength during the focal windows; the bootstrap tests how unusual the sample means are within the observed return history.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

	Mean-adjusted		Peer-adjusted	
Event	React $[0,1]$	Drift $[2,10]$	React $[0,1]$	Drift $[2,10]$
GLM-5	+22.5	+24.7	+17.2	+13.3
GLM-5-Turbo	+7.7	−19.3	+14.6	+19.2
GLM-5.1	+13.8	−14.2	+13.4	+12.9
GLM-5.2	+30.8	+28.2	+28.7	+36.8
GLM-5.3	−6.5	+3.8	+2.4	+1.8
Average	+13.7	+4.6	+15.3	+16.8

Mean-adjusted uses Zhipu's pre-event average return over $[-20,-6]$; peer-adjusted uses MiniMax's same-day return as the benchmark. All five GLM drift windows are complete through +10.

5.4 Block-bootstrap evidence and limits

With five focal events and volatile returns, inference depends heavily on the sampling design. We draw 20,000 bootstrap samples of five peer-adjusted return blocks, matching the two-day reaction and nine-day drift lengths. Eligible blocks lie outside the five focal GLM $[0,+10]$ windows; they can still contain other financial, product or flow news. The comparison therefore uses the observed trading environment as a reference, not a news-free counterfactual (Figure 11).

The actual mean reaction of +15.3% exceeds the 95% null interval of $[-14.0,+12.6]$% and lies at the 99.2nd percentile. Mean drift of +16.8% lies inside the wider $[-22.8,+25.5]$% interval, at percentile 88.0. Immediate relative responses are unusually large under this design; evidence for persistent drift is inconclusive. This result does not isolate a causal capability effect, estimate an AGI-success probability or establish a general market-efficiency anomaly. A larger multi-laboratory sample and independently scored surprise magnitudes would be needed for those stronger questions.

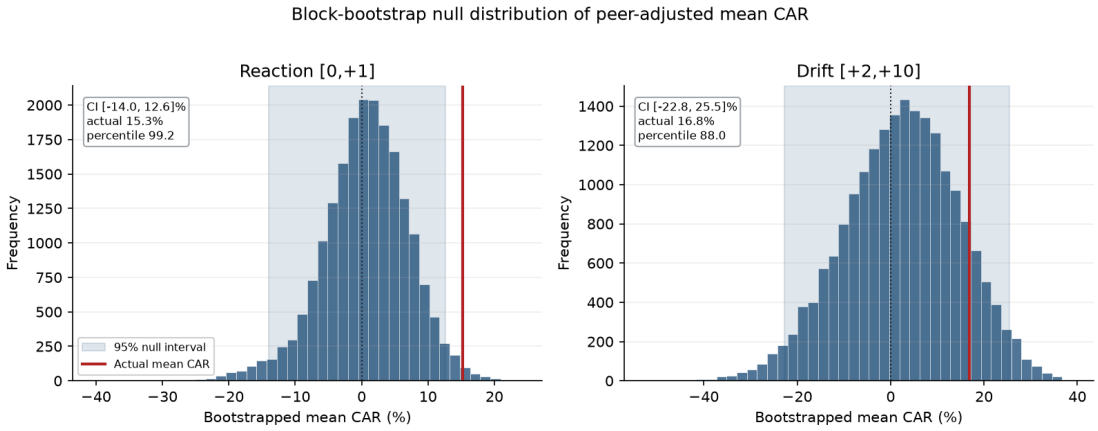


Figure 11: Block-bootstrap null distributions for peer-adjusted mean CAR. Each draw samples five return blocks outside the focal GLM windows, with replacement; block lengths match the actual event windows. Shaded areas are percentile 95% null intervals, and red lines mark the actual five-event means.

6 September Repricing and Commercial Conversion

6.1 Complete supplementary windows

The September extension provides observations that were incomplete at the original August cutoff. GLM-5.3-Flash records a +9.1% reaction followed by −25.9% drift. Zhipu's H1 announcement, mapped to 1 September after its 31 August after-close release, records −8.1% reaction and −54.4% drift. MiniMax's H1 announcement records −5.7% and −23.2%. The Tencent Hy4 and MSCI windows are also complete. These observations remain outside the fixed five-GLM estimates; all are recorded in `eventstudy/september_window_update.csv`.

The Flash, financial-results and index windows overlap. Their negative subsequent CARs describe the repricing episode but are not independent replications of a capability-related reversal. The two companies' financial windows also differ in timing. They cannot identify how much of the decline came from margins, funding expectations, sector demand or other concurrent information.

Over the full calendar period from 31 August to 30 September, Zhipu fell 46.36% and MiniMax 28.31%, a difference of 18.05 percentage points. These are compounded raw-price returns, distinct from event CARs. Relative performance also depends on the selected period: from 24 to 30 September, Zhipu gained 0.39% while MiniMax fell 10.32%, giving a 10.71-percentage-point spread. The month-end improvement shows that the relative repricing was not monotonic.

The decline represents a partial unwinding of earlier gains. Interpreting it as a compression of optimistic commercial expectations is consistent with the remaining 191.3× LTM multiple and the conditional DCF gap, but these methods do not estimate a capability premium as a separate price component. We do not estimate a long-run price mean, test stationarity or demonstrate convergence to DCF value. Commercial costs and financing requirements offer plausible explanations for the changing valuation; overlapping news prevents causal attribution.

6.2 Developer trust, tooling and incomplete windows

On 18 September, developer ferstar alleged that ZCode uploaded workspace snapshots including Git history and that settings did not disable the upload path[55]. The analysis was not independently reproduced here. In an evening response reported by IT Home, ZCode apologised, attributed uploads to default-enabled indexing or Repo Wiki, said the issue was fixed and data destroyed after Wiki generation, and promised source release and external review[56]. These statements are company claims. ZCode published Apache-2.0 source on 21 September, followed by v3.14.3 on 23 September[60]. It reported removal of Repo Wiki and the snapshot-upload path in v3.14.0, citing CAICT and NSFOCUS reviews[61]. The signed review reports and historical deletion evidence were not independently verified. Available evidence does not establish that every repository was uploaded or used for training.

MiniMax Code CLI v0.4.12 was released under MIT as an inspectable and extensible terminal coding-agent implementation[58]. It differs from the older general-purpose MiniMax CLI and from model weights. The two developments illustrate competition for the developer workflow: ZCode seeks to restore confidence in data handling, while public CLI source can make an alternative easier to evaluate. Current source alone does not verify historical or deployed behaviour, and the evidence does not show that MiniMax timed its release in response to ZCode.

Data-control confidence can affect enterprise approval, developer trials, retention and paid usage. Remediation costs or longer procurement cycles could affect Zhipu; MiniMax could gain consideration without capturing revenue. Its CLI supports other models, so tool adoption need not increase MiniMax API consumption. No disclosed migration or revenue-loss estimate sup-

ports changing the operating DCF assumptions. The mechanism matters to valuation even while its quantitative effect remains unknown.

Timing limits the supplementary return evidence. Allegations circulated during 18 September trading; evening responses and CLI coverage followed[56–58]. Mapping that later information to 21 September leaves two reaction days and six of nine drift days at the cutoff. The formal interim report released after the 24 September close maps to 25 September, with two reaction days and two drift days. Reported weekend preview availability maps to 28 September, leaving two reaction days and one drift day; the earliest availability timestamp is not independently established. MiniMax's 29 September notice confirms M3.1-Flash-Preview and forthcoming subscription migration[65]. October offers remain prospective. These partial windows are reported separately and cannot support full-window comparisons.

6.3 Capital access and uncertain upside

A high equity valuation can support research by allowing a firm to raise cash for compute and talent. This financing feedback resembles the reflexivity interpretation discussed by Feng Liu[49]. July's placement at HK$1,588 and September's at HK$714 illustrate how financing terms can change as prices move, although their dates, sizes and terms differ. New funding extends investment capacity, while dilution and debt repayment affect how much eventual success accrues to each share. Access to capital is therefore part of the commercial conversion process, not evidence that the process has already succeeded.

Real-options theory offers an analogy for uncertain upside[47,48]: investors may value opportunities to expand into a much larger market. We do not calculate an option price or infer a unique probability from the gap above DCF. Different growth expectations, risk assessments and model omissions can produce the same gap. The event study measures short-run returns around announcements; it does not observe changes in the perceived probability of an AGI outcome. The reverse DCF makes the required operating scale explicit without choosing between these interpretations.

6.4 Anthropic as a comparison of valuation methods

The Anthropic comparison examines how a large revenue forecast translates into shareholder value once operating costs, investment and discounting are specified. Information is restricted to 30 September 2026, while the model assumes a 31 December 2026 valuation date. Anthropic confirmed a confidential S-1 submission; Reuters reported an approximately US$2tn IPO equity aspiration and a US$190–200bn revenue projection for 2028[62–64]. The completed May funding transaction remains the observation used in the private-company comparable set[42]. Table 4 separates these observations from the assumptions used below.

Table 4: Anthropic comparison: reported information and model assumptions. Dollar amounts are US$bn.

Item	Amount	Status and valuation use
May funding valuation	965	Completed transaction, reported post-money equity value.
May revenue run-rate	> 47	Company-reported annualised operating pace; the existing comparable uses 965/47, or 20.5×.
IPO equity aspiration	$\approx 2{,}000$	Reuters-reported target; no completed IPO valuation.
2026 revenue anchor	60	Analyst assumption, not company annual guidance.
2028 revenue	195	Midpoint of the reported 190–200 forecast range; used to calibrate the projection, not audited revenue.
Forward EV/revenue	10×	Assumed multiple applied to 2028 revenue, producing US$1,950bn EV.

The DCF projects ten year-end unlevered free cash flows (UFCFs) over 2027–2036, followed by terminal cash flow from 2037. Revenue doubles to US$120bn in 2027 and grows 62.5% to US$195bn in 2028. Growth then declines to 3% in 2036, when revenue reaches US$578.0bn. EBIT margin increases from 10% to 35%, and positive EBIT incurs 25% cash tax. The model assumes D&A at 3% and capex at 8% of revenue. Their net cash investment cost is therefore 5% of revenue before working capital.

Working capital is a balance equal to 2% of revenue. Each year's UFCF deducts the increase in that balance, rather than a recurring charge equal to 2% of revenue. Starting working capital is US$1.2bn on the assumed US$60bn 2026 revenue anchor. In 2028, for example, the US$75bn revenue increase requires US$1.5bn of additional working capital. The cash-flow bridge is

$$\mathrm{UFCF}_t = \mathrm{EBIT}_t(1 - 0.25) + \mathrm{D\&A}_t - \mathrm{Capex}_t - \Delta\mathrm{NWC}_t.$$

Table 5 shows selected years from the annual projection. The US$195bn revenue forecast yields US$18bn of 2028 UFCF under these assumptions.

Table 5: Selected Anthropic cash-flow projections (US$bn, except margin).

Year	Revenue	EBIT margin	After-tax EBIT	Capex − D&A	ΔNWC	UFCF
2027	120.00	10%	9.00	6.00	1.20	1.80
2028	195.00	20%	29.25	9.75	1.50	18.00
2030	329.06	30%	74.04	16.45	1.32	56.27
2036	578.04	35%	151.74	28.90	0.34	122.50

At 12% WACC and 3% terminal growth, the explicit cash flows have a present value of US$336.9bn and the terminal value contributes US$451.4bn. Total enterprise value is US$788.3bn, with 57.3% attributable to terminal present value. Table 6 varies WACC and perpetual growth while retaining the explicit operating path and terminal investment ratios. The resulting EVs range from US$581.7bn to US$1,197.1bn within this grid. This sensitivity describes the specified model; it does not cover changes in revenue, margins or investment intensity.

Table 6: Anthropic DCF sensitivity: enterprise value (US$bn).

WACC	$g = 2\%$	$g = 3\%$	$g = 4\%$
10%	981.9	1,074.1	1,197.1
12%	739.6	788.3	849.2
14%	581.7	609.9	643.9

A reverse calculation shows what the US$2tn benchmark requires if the explicit cash flows, terminal revenue, tax, investment ratios, WACC and terminal growth are held fixed. Changing only the terminal EBIT margin gives a required margin of 110.9%, approximately 111%. This is a conditional diagnostic: the fixed revenue and investment path cannot support the benchmark through an ordinary operating margin alone. It is not a literal forecast or evidence that every US$2tn valuation scenario is impossible. A different revenue path, earlier margin improvement, lower investment requirements or a lower discount rate would change the constraint.

The treatment of compute costs also affects the comparison. Reuters reports cloud-distribution dependence and substantial long-term compute commitments[64]. Payments for rented compute, training and distribution may enter operating expenses and therefore EBIT margin; owned infrastructure may require capex. The same cost should not be deducted in both places. Conversely, fixed margin and capex ratios may understate the timing or scale of committed expenditure. The model does not contain a contract-level expenditure schedule, so it cannot establish that these ratios fully capture the commitments. Long-term compute commitments are not automatically current interest-bearing debt.

Net cash is assumed zero, temporarily equating enterprise and equity value. No IPO proceeds, new debt or fully diluted share count are incorporated because their amounts are not established in this comparison; no per-share IPO price is estimated. The assumed 10× forward revenue multiple also uses a different denominator from the May funding reference's 20.5× annualised-revenue multiple, so the difference does not measure observed multiple compression. Drivers, annual cash flows and sensitivities are reproduced in `scripts/anthropic_ipo_crosscheck.py` and its CSV outputs.

For Zhipu, this comparison clarifies why a revenue forecast alone cannot determine shareholder value. The Anthropic and Zhipu models have different valuation dates, taxes, discount rates and reinvestment methods, so their absolute values are not a controlled comparison or an additional Zhipu price target. In each model, valuation depends on how revenue becomes operating profit, how much investment growth requires and how financing affects the claim held by shareholders.

7 Conclusion and Implications

The five original GLM release windows show positive average immediate returns, including an unusually large mean response relative to MiniMax under the block-bootstrap design. Evidence for persistent drift is weaker: it remains inside the bootstrap null interval, and individual signs change with the benchmark. September's decline shows that the earlier re-rating did not establish a durable price floor. The sample does not identify a capability-driven pricing error or statistical mean reversion.

The operating evidence makes commercial conversion the central valuation question. H1 revenue grew 399.7% and API gross margin turned positive, while operating loss exceeded twice revenue and cash outflow increased. Against the HK$641 close, scenario-weighted DCF gives HK$116, the illustrative private-transaction range HK$230–437, and the matched MiniMax LTM multiple HK$227. The reverse DCF requires approximately US$47.9bn of revenue in 2035 under its selected assumptions. These conditional gaps describe how much growth and operating improvement the price requires; they do not provide a proven reversion target.

Financing capacity, developer trust and customer retention connect model quality to future cash flow. Completed placements and convertible issuance provide resources alongside dilution and repayment obligations. ZCode's remediation and competing inspectable tools may affect supplier evaluation, but migration and revenue effects remain unmeasured. The Anthropic comparison reinforces the need to specify margins and investment requirements even when revenue expectations are very large. Further evidence on retained paid usage, unit economics, cash consumption and financing terms would determine whether technical progress becomes durable shareholder value.

A Financial Statements Summary

From the Chapter 18C prospectus, 2025 annual report and H1 2026 interim-results announcement (RMB; USD at ≈7.1).

Table 7: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	FY2025
Revenue	57.4	124.5	312.4	724.3
Total gross margin	54.6%	64.6%	56.3%	41.0%
Cloud gross margin	76.1%	31.0%	3.4%	18.9%
Net loss	143.7	788.0	2,958.0	4,718.2

Table 8: H1 operating scorecard (RMB millions unless noted).

	H1 2025	H1 2026	Change
Revenue	190.9	953.9	+399.7%
Open-platform and API revenue	29.1	825.2	+2,735.7%
Total gross margin	50.0%	26.4%	−23.6pp
API gross margin	−0.4%	24.6%	+25.0pp
Operating loss	1,899.2	2,146.6	+13.0%
Adjusted net loss	1,752.0	1,964.1	+12.1%

Table 9: Balance-sheet and placement bridge (RMB millions at 30 June 2026 unless noted).

Item	Amount
Bank cash (including restricted deposits)	3,993.7
Short-term investments at FVPL	506.1
Bank loans	2,224.8
Lease liabilities	379.4
Total equity	4,429.6
Shares outstanding after September placement	487.588m
Placement price / net proceeds	HK$1,588 / HK$31,375.0m
September placement net proceeds	HK$15,664.1m
CB net proceeds / principal proxy	US$3,010.8m / US$3,000.6m
Model pro-forma net cash and investments	US$6,292.9m

The formal interim report adds the cash-flow statement (pp. 60–61). H1 operating, investing and financing flows are RMB−3,374.057m, RMB−702.725m and RMB+5,785.942m. Opening cash equivalents of RMB2,254.920m plus these flows and RMB−76.353m of FX effects reconcile to RMB3,887.727m. Note 14 deducts RMB105.995m restricted deposits from bank cash. Accordingly, the model excludes US$14.929m from its net-cash bridge, lowering each DCF per-share value by HK$0.239. The revised bridge is US$6,292.850m. H1 cash burn is already reflected in June cash and is not deducted again. Subsequent cash consumption remains unreported[59].

B Base-Case DCF Projection

Base-case chain (US$m unless noted), reproducing the workbook: discount rate = 13.5%, model tax assumption = 15%, terminal g = 4%, sales-to-capital = 1.8, 487.588m shares and HK$7.8/US$. Revenue begins at US$700m FY2026E and reaches US$8.0bn in 2035, with growth fading from 56% in 2027 to 8% in 2035. Operating margin rises from −100% to 28%. Negative EBIT accumulates NOLs, which shelter later positive EBIT in this simplified forecast.

The Base case still has US$4.369bn of NOLs at the end of 2035. Terminal value extrapolates the final forecast FCFF at 4% growth; it does not separately normalise cash taxes after the loss balance is exhausted. It also carries forward reinvestment based on the final explicit-year 8% revenue growth rather than recalculating it at 4% terminal growth. These conventions have opposing valuation effects: zero-tax extrapolation tends to raise value, while retaining the higher investment intensity tends to reduce it. A normalised terminal forecast requires explicit post-2035 tax and reinvestment assumptions. The DCF and reverse DCF remain conditional on this end-year cash-flow convention.

Table 10: Base-case projection (US$ millions unless noted).

Year	Revenue	Growth	EBIT margin	Cash tax	NOL end	NOPAT	Reinvest	FCFF
2026	700	586%	−100.0%	0	700	−700	332	−1,032
2027	1,092	56%	−85.8%	0	1,637	−937	218	−1,155
2028	1,638	50%	−71.6%	0	2,809	−1,172	303	−1,475
2029	2,359	44%	−57.3%	0	4,161	−1,352	400	−1,753
2030	3,255	38%	−43.1%	0	5,564	−1,403	498	−1,901
2031	4,297	32%	−28.9%	0	6,806	−1,241	579	−1,820
2032	5,414	26%	−14.7%	0	7,600	−794	621	−1,415
2033	6,497	20%	−0.4%	0	7,629	−29	602	−630
2034	7,406	14%	13.8%	0	6,608	1,020	505	515
2035	7,999	8%	28.0%	0	4,369	2,240	329	1,910

Terminal value (2035) = $1{,}910 \times 1.04/(0.135 - 0.04)$ = US$20,914m; PV of TV = US$5,895m.
Enterprise value = US$55m; + pro-forma net cash and investments US$6,293m = equity US$6,348m = **HK$102/share**.

C Bottom-Up Beta Bridge

The bottom-up beta uses public software and AI-infrastructure comparables with observable market betas and capital structures. Levered betas use 60 month-end returns through 31 August 2026; market value is that day's close times the latest disclosed shares, and debt comes from the latest pre-cutoff SEC filing. We unlever as $\beta_U = \beta_L/[1+(1-T)D/E]$, using a 21% tax rate, market capitalisation as equity value, and total debt as debt[40]. We round the median unlevered beta of 1.55 to 1.6 as a working reference for operating risk. No Zhipu-specific re-levering or debt-and-equity-weighted WACC is calculated; 13.5% remains the common discount-rate assumption used by the valuation scenarios. WACC labels in the model tables and figures refer to this discount-rate assumption.

Table 11: Comparable-company beta bridge (60-month beta window ended 31 August 2026).

Comparable	Levered beta	Debt/equity	Unlevered beta
Salesforce	1.19	20.0%	1.03
Palantir	1.61	0.0%	1.61
Snowflake	1.33	2.4%	1.30
Datadog	1.48	1.5%	1.46
MongoDB	1.55	0.2%	1.55
Cloudflare	1.66	4.4%	1.60
C3.ai	2.11	3.4%	2.05
Median	—	—	**1.55**

D Reverse-DCF Sensitivity Grid

Table 12 reports the 2035 revenue (US$bn) required to justify the US$40.1bn equity value across a grid of WACC and terminal-margin assumptions, underlying Figure 3.

Table 12: Reverse-DCF sensitivity: required 2035 revenue (US$bn) to justify US$40.1bn equity.

WACC	Terminal operating margin				
	20%	28%	35%	40%	50%
10.0%	60	30	21	18	14
11.5%	102	47	33	28	22
13.0%	170	71	51	42	30
13.5%	202	82	58	48	34
15.0%	339	127	84	68	46
17.0%	703	223	135	103	66
20.0%	4284	524	264	183	110

Base-case WACC is 13.5%. All figures hold FY2026E revenue at US$700m and solve the 2027 growth rate, after which growth fades linearly to 8% in 2035. They also assume $g = 4\%$, sales-to-capital 1.8 and a 15% tax rate with NOL carryforwards.

E Wenge: Adjacent-Comparable Financial Detail

Wenge's 25 August interim-results announcement reports H1 revenue of RMB172.750m, up 46.0%. Deployment contributes RMB113.803m (65.9%), while subscription and API revenue reaches RMB58.947m (34.1%), up 94.0%. Gross margin is 55.0%, versus 54.6% a year earlier. R&D expense rises 59.0% to RMB119.340m, or 69.1% of revenue[53]. Its deployment-heavy mix differs from Zhipu's API business.

Total period loss narrows 24.5% to RMB87.654m; loss attributable to owners narrows 27.6% to RMB88.958m; adjusted net loss narrows 29.5% to RMB52.920m. Financial-asset fair-value gains increase from RMB0.200m to RMB35.047m, contributing to the improvement in reported losses. Cash and cash equivalents increase from RMB324.621m at year-end to RMB1,017.608m in June. The broader liquidity increase is primarily attributable to IPO proceeds and does not demonstrate positive operating cash flow. Revenue growth and adjusted-loss improvement occur alongside faster R&D growth.

The comparable table uses an H1-updated LTM denominator. The source is the formal interim-results announcement; a full interim report was not separately verified here. Wenge remains outside the private foundation-laboratory transaction range.

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